

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 3 Maths — CA-1 (Medium) — Answer Key

Class : 3

Subject : Maths (Math-Mela)

Max Marks : 70

Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — What's in a Name?, Toy Joy, Double Century

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. If you group 23 pebbles into groups of 5, how many full groups do you get and how many are left over? (1 Mark)

A. 4 groups, 3 left over ✓

B. 5 groups, 0 left over

C. 4 groups, 5 left over

D. 3 groups, 8 left over

2. How many edges does a cube have? (1 Mark)

A. 6

B. 8

C. 12 ✓

D. 10

3. Which of these shapes can roll AND stack (has both flat and curved surfaces)? (1 Mark)

A. Sphere

B. Cylinder ✓

C. Cube

D. Cuboid

4. A cuboid has how many rectangular faces in total? (1 Mark)

A. 4

B. 6 ✓

C. 8

D. 12

5. What is the value of the digit 9 in the number 198? (1 Mark)

A. 9

B. 90 ✓

C. 900

D. 19

6. Which number is exactly halfway between 100 and 200? (1 Mark)

A. 110

B. 125

C. 150 ✓

D. 175

7. If you write numbers from 190 to 200 in order, how many numbers will you write in total? (1 Mark)

A. 9

B. 10

C. 11 ✓

D. 12

8. Which of these is the correct expanded form of 163? (1 Mark)

A. $100 + 60 + 3$ ✓

B. $100 + 6 + 3$

C. $160 + 3 + 0$

D. $16 + 3$

9. Two shapes together make a toy train engine: a cuboid body and a cylinder chimney. Which property lets the chimney stand upright? (1 Mark)

A. Its flat circular base ✓

B. Its curved surface only

C. It has no faces

D. It can roll easily

SECTION B — Short Answer Questions (2 Marks each)

1. A farmer keeps one small stone in a bag for every sheep that goes out to graze, and takes one stone out for every sheep that returns. How does this help the farmer? (0 Marks)

► If any stones are left in the bag at the end of the day, the farmer knows that many sheep have not returned yet, even without counting big numbers directly.

2. Explain why a cube is a special type of cuboid. (0 Marks)

► A cuboid has rectangular faces that can be different sizes, but when all of a cuboid's faces are equal squares, it becomes a cube - so every cube is a cuboid, but not every cuboid is a cube.

3. Name two objects that combine more than one 3D shape, and describe the shapes used. (0 Marks)

► An ice cream cone with a scoop combines a cone (the holder) and a sphere (the ice cream ball). A toy rocket can combine a cylinder (the body) and a cone (the tip).

4. Write the number that is 10 more than 175, and the number that is 10 less than 175. (0 Marks)

► 10 more than 175 is 185, and 10 less than 175 is 165.

5. Compare 3-digit numbers 142 and 124 using $<$ or $>$ and explain how you decided. (0 Marks)
- $142 > 124$, because although both start with 1 hundred, 142 has 4 tens while 124 has only 2 tens, and more tens makes the number greater.
6. A box contains 100 pencils. If 37 pencils are taken out, how many are left in the box? (0 Marks)
- $100 - 37 = 63$ pencils are left in the box.
7. Write any three numbers between 100 and 200 that have 5 in the tens place. (0 Marks)
- Examples: 150, 152, 158 (any numbers of the form 1_5_ with 5 in the tens place).
8. Explain how counting in groups of 10 makes counting a large pile of buttons easier than counting one by one. (0 Marks)
- Grouping into 10s lets us count fewer, bigger groups (like counting groups instead of every single button), and then just add any leftover ones, which is faster and reduces mistakes.
9. Name a shape used for the wheels of a toy car and explain why that shape is chosen. (0 Marks)
- Wheels are usually cylinder-shaped (like thick circles) because the curved surface lets them roll smoothly in one direction.
10. If a cuboid box has a length longer than its height and width, how would it look different from a cube-shaped box? (0 Marks)
- A cuboid box like this would look stretched out or rectangular, while a cube-shaped box would look perfectly even on all sides, like a dice.
11. Write the smallest and largest 3-digit numbers you can make using the digits 1, 8, and 4 (using each digit once). (0 Marks)
- Smallest: 148. Largest: 841.
12. A shopkeeper arranges 120 sweets into jars of 20 sweets each. How many jars are needed? (0 Marks)
- $120 \div 20 = 6$ jars are needed.
13. How are a cone and a cylinder similar, and how are they different? (0 Marks)
- Both have a curved surface and at least one flat circular face; but a cylinder has two flat circular faces of the same size, while a cone has only one flat circular face and comes to a point at the top.
14. Write in words: 200. (0 Marks)
- Two hundred.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. A group of children want to count 156 marbles without losing track. Describe a step-by-step method they could use, connecting it to grouping in tens and hundreds. (0 Marks)
- First, make small groups of 10 marbles each until no more full groups of 10 can be made.
 - Then, group ten of those groups of 10 together to make a group of 100, if there are enough marbles.
 - Count the leftover single marbles at the end.
 - For 156 marbles: 1 group of 100, 5 groups of 10, and 6 single marbles, giving $100 + 50 + 6 = 156$.
2. Explain how the properties of shapes (flat/curved surfaces, corners) help decide whether an object can roll, stack, or slide. (0 Marks)
- Shapes with only curved surfaces and no flat faces, like a sphere, can roll in any direction easily.
 - Shapes with flat faces, like cubes and cuboids, can be stacked on top of each other because they sit steadily without rolling.
 - Shapes like cylinders and cones have both flat and curved parts, so they can roll on their curved side but also stand still on their flat face.
3. Three numbers are given: 187, 178, 168. Write them in decreasing order and explain your method of comparing. (0 Marks)
- To compare, look at the hundreds digit first; all three numbers have 1 hundred, so we compare the tens digit next.
 - 187 has 8 tens, 178 has 7 tens, and 168 has 6 tens, so 187 is the greatest and 168 is the smallest.
 - Decreasing order: 187, 178, 168.
4. Design a simple toy using at least three different 3D shapes. Describe which shapes you used and where. (0 Marks)
- A toy house could use a cuboid for the main body of the house, a cone for the roof, and small cylinders for chimneys.
 - The flat faces of the cuboid let it sit steadily on the ground, while the cone shape gives the roof its pointed look.
 - This shows how different shapes can be combined to build something new and useful.
5. A basket has 200 apples. 45 are given away, and then 30 more are sold. How many apples remain? Show each step. (0 Marks)
- Start with 200 apples.
 - After giving away 45: $200 - 45 = 155$ apples remain.
 - After selling 30 more: $155 - 30 = 125$ apples remain.
6. Explain why understanding place value (ones, tens, hundreds) is important when comparing numbers like 199 and 200. (0 Marks)
- Place value tells us the actual worth of each digit based on its position - ones, tens, or hundreds.
 - 199 has 1 hundred, 9 tens, and 9 ones, while 200 has 2 hundreds, 0 tens, and 0 ones.
 - Even though 199 has bigger-looking digits, 200 is greater because it has more hundreds, showing why place value matters more than just the digits themselves.

SECTION D — Long Answer Questions (5 Marks each)

1. Explain, with examples, how the ideas in "What's in a Name?" (counting and tracking) connect to the ideas in "Double Century" (numbers up to 200) as a child learns to count larger and larger amounts. (0 Marks)

- In "What's in a Name?", children learn basic ways to track quantities, like using tally marks or one-to-one matching, which works well for small numbers.
- As amounts grow bigger, like up to 200, it becomes hard to track every single item one at a time, so grouping into 10s and 100s (place value) becomes necessary, which is the focus of "Double Century."
- For example, tracking 8 cows can be done with 8 tally marks, but tracking 180 sheep is much easier by counting in groups of 10 and 100 instead.
- This shows how early counting skills build the foundation for understanding bigger numbers and place value later.

2. Using shapes learnt in "Toy Joy," design and describe (in words) a toy shop display that uses cubes, cuboids, cylinders, cones, and spheres, explaining why each shape suits its use. (0 Marks)

- Building blocks in the shop could be cubes, since their equal flat faces let children stack them easily into towers.
- Toy boxes for storage could be cuboids, since their rectangular shape uses shelf space efficiently.
- A toy drum could be a cylinder, since its flat circular ends and curved side make a good drum shape.
- Party hats could be cones, since the pointed shape sitting on a flat circular base is a classic party hat design.
- Bouncy balls could be spheres, since their fully curved surface with no flat faces lets them roll and bounce in every direction.

3. A shopkeeper starts the day with 200 toys. Throughout the day, 68 toys are sold in the morning and 47 more in the afternoon, but 15 toys are returned by customers in the evening. How many toys does the shopkeeper have at the end of the day? Show all your working step by step. (0 Marks)

- Start with 200 toys.
- After morning sales: $200 - 68 = 132$ toys.
- After afternoon sales: $132 - 47 = 85$ toys.
- After evening returns (adding back): $85 + 15 = 100$ toys.
- The shopkeeper has 100 toys at the end of the day.